

Lab Manual for Digital Logic and Design

LAB-05

Implementation of the given Boolean function using logic gates in both SOP and POS forms

EXPERIMENT

NO.: 05

Implementation of the given Boolean function using logic gates in both SOP and POS forms

Objective:

To implement the given Boolean function using logic gates in both SOP and POS forms

Two input SOP - $A.B + A'.B'$

Three input POS: - $(A+B)(B+C)(A+C')$

Apparatus:

- Logic trainer
- Connecting wires
- 14 pin ICs (7408,7432,7404)0
- Power supply

Theory:

SOP: - It is the Sum of product form in which the terms are taken as 1. It is denoted in the K-map expression by sigma (Σ)

A.B. + A'B'

Logic Circuit of this expression: -

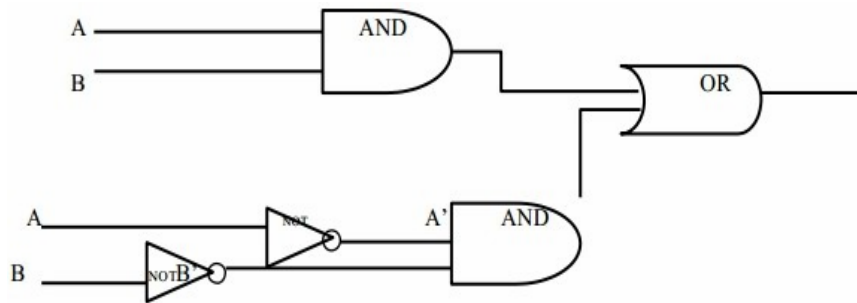


Figure. 5.1: Logic Diagram of SOP form

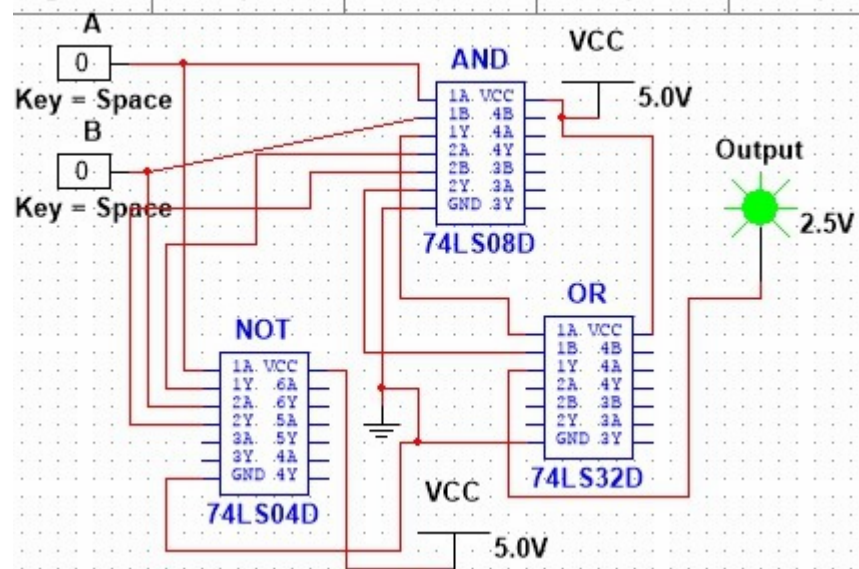


Figure. 5.1.1: Simulation of SOP form

POS: - It is the product of the sums form in which the terms are taken as 0. It is denoted in the K-Map expression by the Sign pie (π)

$$(A+B)(B+C)(A+C')$$

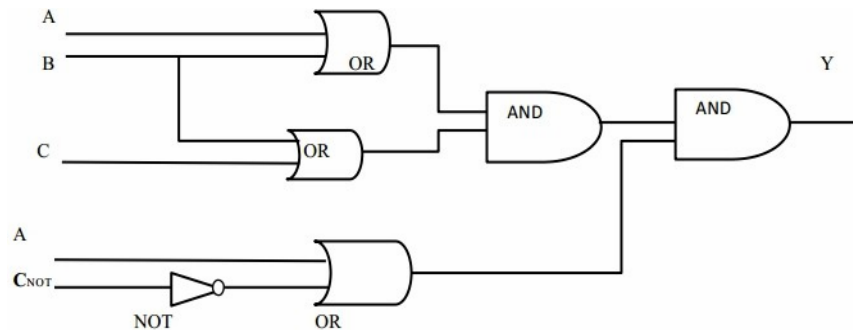


Figure. 5.2: Logic Diagram of POS form

Simulation:

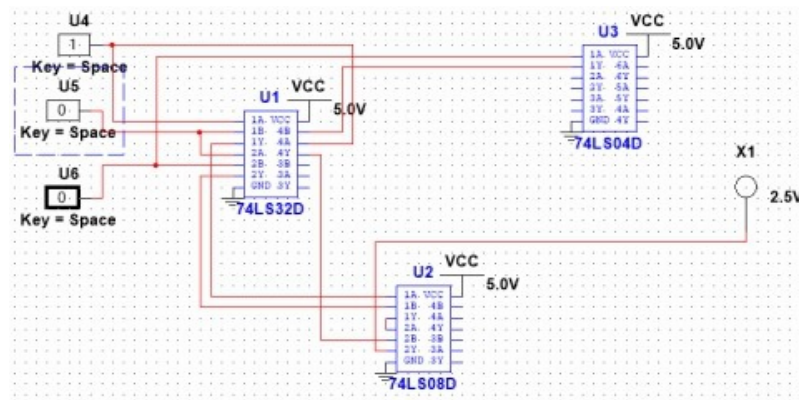


Figure. 5.2.1: Simulation of SOP form

Part (b)

Observations & Calculations:

Table 5.1: Truth Table for the SOP expression:

A	B	A'	B'	A.B	A'B'	Y=AB+A'B'
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1

Table 5.2: Truth Table for the POS expression:

A	B	C	A+B	B+C	A+C'	Y=(A+B)(B+C)(A+C')
0	0	0	0	0	1	0
0	0	1	0	1	0	0
0	1	0	1	1	1	1
0	1	1	1	1	0	0

1	0	0	1	0	1	0
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Procedure:

- Construct the circuits shown in logic diagrams.
- Connect inputs to switches of logic trainer and output to LED.
- Apply various combinations of inputs and observe the conditions of LEDs.
- Mark results in truth table.

Questions:**1) What is a combinational circuit?**

Ans: Combinational Circuit is the type of **circuit** in which output is independent of time **and** only relies on the input present at that particular instant.

2) What is a sequential circuit?

Ans: Sequential circuit is the type of **circuit** where output not only relies on the current input but also depends on the previous output.

Conclusion:

Implement the following circuits in MultiSim. The outputs were observed on LED's and outputs were same as truth table.